

MOP Web Application

Local Environment Configuration Audit & Production Environment Preparation

1. Objective

The objective of this task was to review the current environment configuration files (.env and .env.example), identify all environment variables required by the application, and prepare a production-ready environment configuration for Sprint 2.

As part of the project's deployment strategy, the application is transitioning from the previous Supabase based backend towards a MongoDB based architecture hosted on Google Cloud Platform (GCP). This audit identifies configuration changes required for the migration and prepares a secure manifest for the Team Leads to configure Google Cloud Secret Manager during production deployment.

2. Files Reviewed

- .env
- .env.example

3. Current Environment Audit

The existing environment configuration contains variables for several services, including Supabase, PostgreSQL, MongoDB, Firebase, SMTP and Sentry.

The project is currently transitioning away from Supabase and local development databases towards a cloud hosted MongoDB deployment on GCP. Therefore, not all existing environment variables will be required in the production environment.

Configuration	Current Status	Sprint 2 Status
Supabase	Used in development	Being removed
PostgreSQL	Local development database	Being replaced by MongoDB
MongoDB	Previously optional	Primary production database
Firebase	Pending verification	Retain only if required
SMTP	Required	Required
JWT	Required	Required
Sentry	Optional	Optional
Google Cloud Secret Manager	Not configured	Required

4. Configuration Changes for Sprint 2

Remove Supabase

The following variables are no longer required once the migration is complete:

- SUPABASE_URL
- SUPABASE_API_KEY

These values should be removed from the production environment after all Supabase dependencies have been migrated.

Replace PostgreSQL

The current project contains PostgreSQL configuration for local development.

These variables should not be used once MongoDB becomes the production database.

Variables to be removed:

- DATABASE_URL
- POSTGRES_HOST
- POSTGRES_PORT
- POSTGRES_DB
- POSTGRES_USER
- POSTGRES_PASSWORD

Configure MongoDB

MongoDB will become the primary production database.

The production environment will require:

- MONGODB_URI

The connection string should point to the production MongoDB instance and must be stored securely using Google Cloud Secret Manager.

Configure Google Cloud Platform

The production deployment will utilise Google Cloud Platform.

Sensitive configuration values should not be stored directly inside .env files on production servers.

Instead, Google Cloud Secret Manager will securely provide secrets to the application during deployment.

5. Production Environment Variables

Required Secrets

Variable	Purpose	Store in Secret Manager
MONGODB_URI	Production MongoDB connection	Yes
JWT_SECRET	JWT signing key	Yes
SMTP_USER	Email account	Yes
SMTP_PASSWORD	SMTP application password	Yes

Required Application Configuration

Variable	Purpose	Secret
SMTP_HOST	SMTP server	No
SMTP_PORT	SMTP port	No
SMTP_FROM	Sender email	No
NEXT_PUBLIC_APP_URL	Production website URL	No

Optional Configuration

These should only be configured if the services remain in use.

Firebase

- NEXT_PUBLIC_FIREBASE_API_KEY
- NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN
- NEXT_PUBLIC_FIREBASE_PROJECT_ID
- NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET
- NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID
- NEXT_PUBLIC_FIREBASE_APP_ID
- NEXT_PUBLIC_FIREBASE_MEASUREMENT_ID

Sentry

- SENTRY_DSN
- NEXT_PUBLIC_SENTRY_DSN

6. Google Cloud Secret Manager Manifest

The following secrets should be created in Google Cloud Secret Manager before deployment.

Secret Name	Description
mongodb-uri	MongoDB production connection string
jwt-secret	JWT signing secret
smtp-user	SMTP email account
smtp-password	SMTP application password

The application should access these values through environment variables during deployment rather than storing credentials directly in source code.

7. Recommendations

- Remove all Supabase configuration after the migration is completed.
- Remove PostgreSQL configuration if MongoDB becomes the sole production database.
- Store all sensitive credentials in Google Cloud Secret Manager.
- Keep `.env.example` as a development template only.
- Never commit production credentials into Git repositories.
- Verify Firebase and Sentry usage before configuring production secrets.